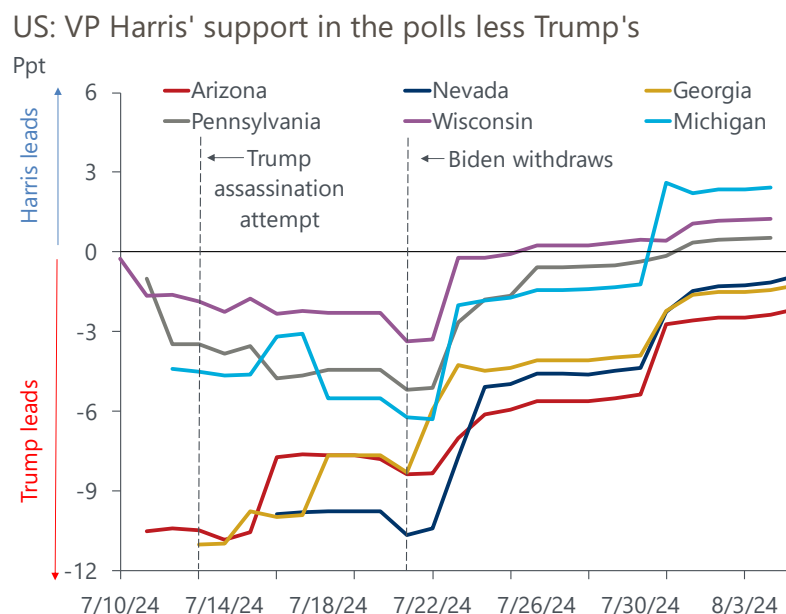


Research Briefing | US

Predicting how swing voters in battleground counties will vote

- The three northern swing states of Pennsylvania, Michigan, and Wisconsin are must-win states for Vice President Kamala Harris and former President Donald Trump. Our presidential election forecast at the county level has the former president projected to win Pennsylvania by 41,489 votes, while we predict the vice president clinches Michigan by 23,948 and Wisconsin by 29,024.
- Pennsylvania is the biggest prize of these three swing states, and we find that county election outcomes in the Keystone State are especially sensitive to changes in unemployment. According to our models, higher joblessness, even if from extremely low levels, dings Harris' support. In 2016, rising unemployment, particularly in the bellwether county of Erie, hinted at trouble for former Secretary of State Hillary Clinton.
- Harris is projected to carry Michigan, but her margin of victory will be quite reduced from Biden's in 2020. President Biden's approval, which is highly correlated with Harris', is a drag on her performance in the Wolverine State, according to our county models. Meanwhile, the vice president is projected to perform the best in Wisconsin, but it is worth noting that the Badger State has a penchant for producing nail-biting election outcomes.
- Harris' selection of Minnesota Governor Tim Walz has limited implications for our county election forecasts. Vice-presidential candidates confer a small home-state advantage to the presidential ticket, and our models don't assume Walz's popularity in Minnesota will spill over into neighboring Wisconsin.

Chart 1: Presidential race tightens in battleground states



Source: Oxford Economics/Silver Bulletin

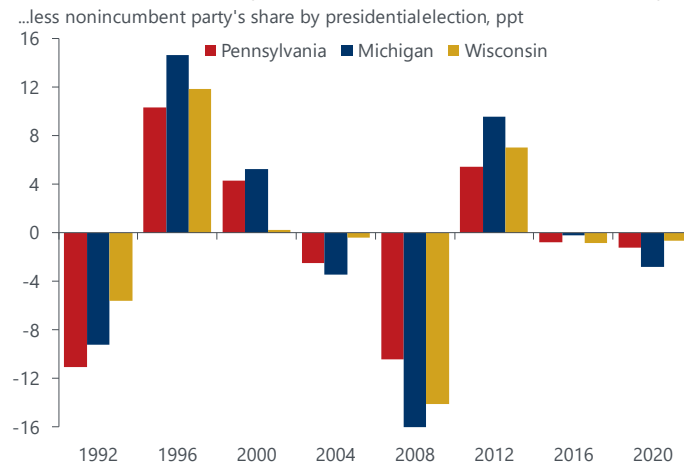
Eight years ago, Pennsylvania, Michigan, and Wisconsin delivered one of the biggest shocks on election night. These previously reliable Democratic states flipped to former President Donald Trump. Pennsylvania and Michigan had not voted for a Republican presidential ticket since 1988, while Wisconsin had steadfastly been in the Democratic column since 1984. In 2020, President Joe Biden won back this trio of northern swing states. This election cycle, the path to victory again cuts through these three battleground states, where the polls show a presidential race that has tightened significantly since [Biden's withdrawal](#) (Chart 1).

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These three states have voted in unison for the same presidential candidate since 1988 (**Chart 2**). As the number of the competitive states has declined, their collective 44 electoral votes have become vital to winning the Electoral College. The latest polls show Harris edging out Trump in Michigan, Wisconsin, and Pennsylvania. However, she still trails the former president in the Sun Belt states of Arizona, Georgia, and Nevada. Therefore, these three northern states represent her most likely path to the White House.

Chart 2: Pennsylvania, Michigan, and Wisconsin are strongly correlated with each other

US: Share of two-party vote for incumbent president's party...



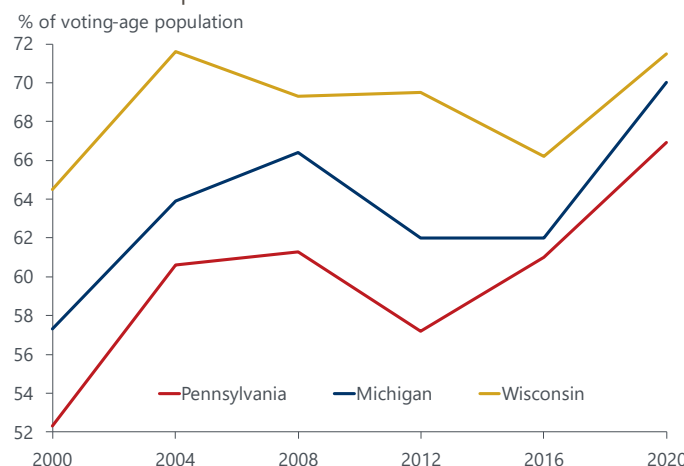
Source: Oxford Economics/Haver Analytics

Using the same methodology as our recently [updated](#) state presidential [election models](#), we created a new set of models that predict the share of the two-party vote for the incumbent president's party at the county level in Pennsylvania, Michigan, and Wisconsin. The former president is projected to win Pennsylvania by a margin of 41,489 votes, while the vice president is predicted to clinch Michigan by a margin of 23,948 and Wisconsin by 29,024 (**Appendix 1**).

Forecasting the margin of victory or defeat for the incumbent president's party in these swing states is a much tougher exercise than predicting the winner of the Electoral College, as our state election models do. Across counties, not only do we have to predict the share of the major-party vote that goes to the incumbent president's party, but also voter turnout (**Chart 3**).

Chart 3: Turnout varies from one election to the next

US: Turnout in presidential elections



Source: Oxford Economics/US Elections Project

However, our county election models have a solid track record in predicting the winners of Pennsylvania, Michigan, and Wisconsin in past presidential elections (**Appendix 2**). While there is a high degree of

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uncertainty around our projections in this unprecedented election cycle, they underscore what a nailbiter this election will be, which dovetails with recent polling trends.

Just like the state election models, the county models assume that swing voters cast their ballots with the state of their local economy in mind. [Appendix 3](#) describes our new granular approach toward forecasting election outcomes, but at a high level, the county models use the following variables to predict the performance of the incumbent president's party:

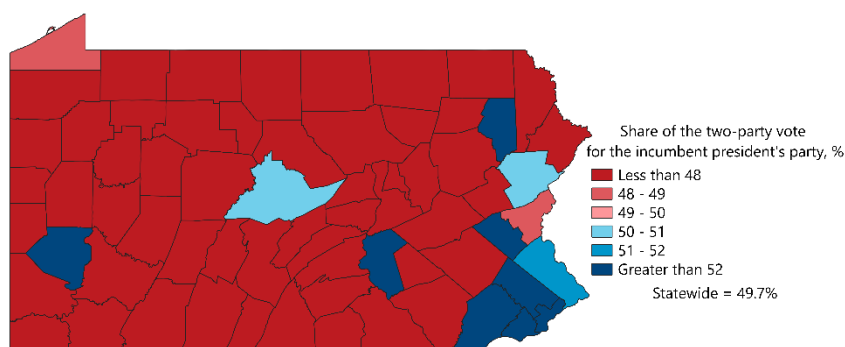
- **Change in county unemployment rates:** Rises in unemployment in the two quarters leading up to the election prompt more swing voters to spurn the incumbent party candidate.
- **County trends in real disposable income per household:** Swing voters, whose financial situations improve due to real, rather than nominal, income gains, will view the incumbent party more favorably.
- **Approval rating of incumbent president:** An incumbent president's popularity helps explain the results of past elections and serves to capture exogenous events that may impact voter behavior.
- **Voter fatigue:** Historically, voters balk at one party sitting in the Oval Office for more than two consecutive terms.
- **Democratic incumbent:** When the incumbent president is a Democrat, Democrats and left-leaning independents are more likely to support the Republican candidate for president than vice versa.
- **Prior vote share:** County fixed effects ensure that reliably blue counties vote for the Democratic candidate and reliably red counties support the Republican candidate.

The value added by the county models is that they highlight the very geographies within these must-win battleground states that will determine the next president.

Pennsylvania is the keystone to an Electoral College win

Of the three northern states, Pennsylvania is the biggest prize with its 19 electoral votes. Our county models suggest Trump will win the Keystone State with 50.3% of the major-party vote ([Chart 4](#)). Even if Harris carries Michigan and Wisconsin, a loss in Pennsylvania would deal her presidential bid a near-fatal blow and would leave her no other option but to win in Georgia and one other swing state in the Sun Belt—a tall order given where things stand in the polls.

Chart 4: How Pennsylvania counties are projected to vote in 2024

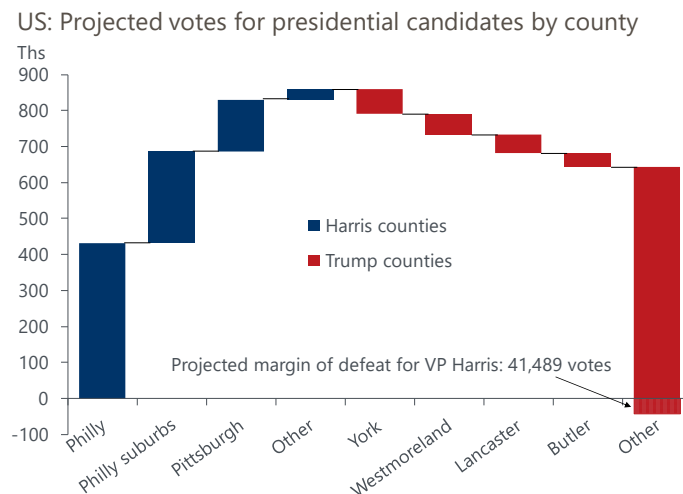


Source: Oxford Economics

The strategy for winning Pennsylvania is straightforward. For Democrats, the goal is to win Philadelphia, the Philadelphia suburbs, and Pittsburgh with the largest possible margins and minimize their losses in the dozens of red counties across the state. Republicans, though, must rack up as many votes as possible outside the state's two largest population centers and limit their margins of defeat in Philadelphia and Pittsburgh. Trump is projected to succeed in this strategy ([Chart 5](#)).

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Chart 5: Harris does not win big enough in Philly and Pittsburgh

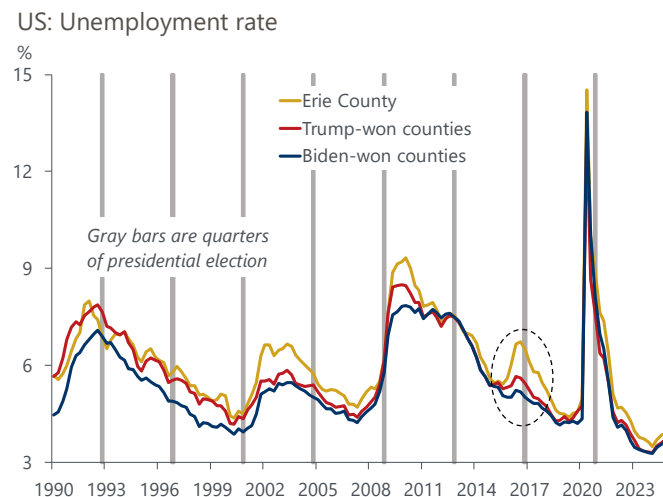


Source: Oxford Economics

One explanation for Harris' insufficient margins of victory in Philadelphia, its suburbs, and Pittsburgh, according to the models, is the unemployment rate. Of the three northern states, county presidential election outcomes in Pennsylvania are statistically the most sensitive to changes in joblessness. Even though the unemployment rate in Pennsylvania is not far from its record low, it's still expected to rise in the quarters leading up to the election, which dings the incumbent party candidate.

In 2016, a rising unemployment rate in Pennsylvania was also a headwind to former Secretary of State Hillary Clinton. At the time, the jobless rate was flat nationally but was inching higher in the Keystone State—and more noticeably so in red counties where Trump generally ended up raking in more votes than prior Republican candidates (**Chart 6**).

Chart 6: Unemployment is rising but from very low levels



Source: Oxford Economics/Haver Analytics

Erie County, which is located on the northwesternmost tip of the state, stood out at the time for its surging unemployment rate due to massive layoffs at GE Transportation. Erie is considered a bellwether county and has voted for the winning presidential candidate of the state without fail throughout the historical sample of our Pennsylvania county models. The county is a microcosm of the state with urban, suburban, and rural areas. Trump is projected to win Erie after Biden narrowly carried the swing county in 2020.

There is one wild card facing our Pennsylvania county election forecasts. The attempted assassination of Trump occurred in Butler County, where the former president is already projected to win one of the largest

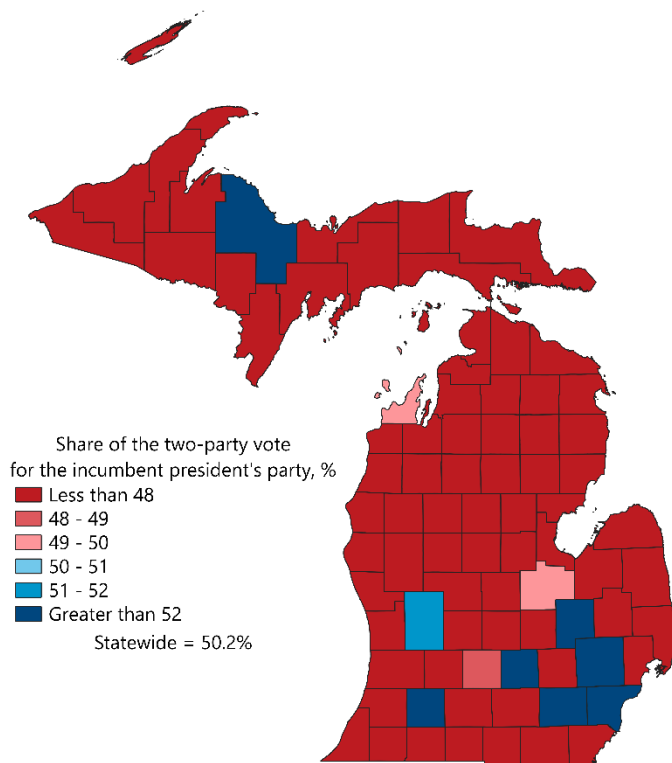
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margins of victory in terms of raw votes. The assassination attempt may galvanize greater Republican turnout than expected—not only in Butler, but also nearby counties such as Westmoreland, Washington, and Beaver that all surround Pittsburgh.

Breaking free from Biden's approval rut is key in Michigan

With 15 electoral votes, our county models in Michigan point to Harris winning the Wolverine State with only 50.2% of the two-party vote (**Chart 7**).

Chart 7: How Michigan counties are projected to vote in 2024

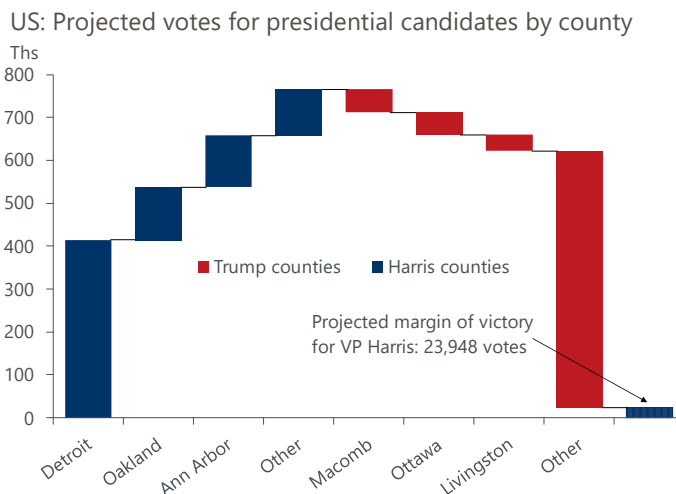


Source: Oxford Economics

For Democrats, electoral success in Michigan means racking up large wins in Detroit and its northern suburbs in Oakland County, as well as Ann Arbor – home to the University of Michigan (**Chart 8**). Meanwhile, Republican candidates must edge out their Democratic rival by as much as possible in the Macomb, Livingston, St. Clair, and Lapeer counties, which surround Detroit and skew whiter and older than the rest of the state, not to mention the dozens of other red counties across Michigan.

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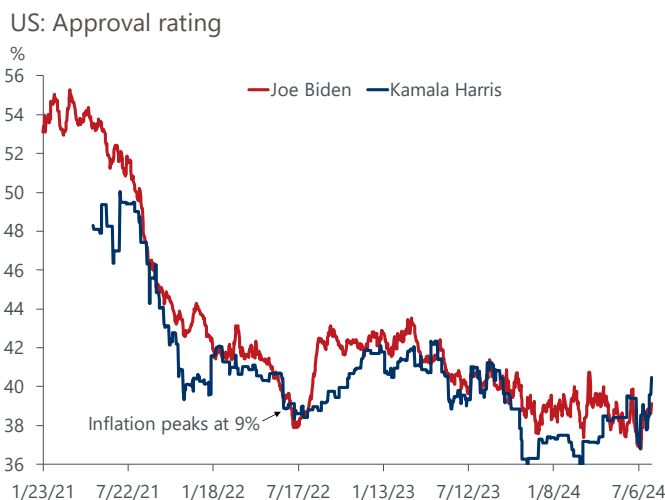
Chart 8: Detroit and Ann Arbor buoy Harris just enough to prevail



Source: Oxford Economics

Though Harris is projected to win Michigan, she does so by a much narrower margin than Biden's 154,188 vote victory in 2020. The variable in our models that holds back her performance is Biden's approval, which is underwater; we find that county presidential election outcomes in Michigan are especially sensitive to swings in the national approval rating of the incumbent president.

Chart 9: Harris' approval moves in tandem with Biden's



Source: Oxford Economics/FiveThirtyEight

It's possible Harris doesn't carry the same political baggage as Biden. However, the data suggests otherwise. Her approval is strongly correlated with Biden's (**Chart 9**). Therefore, using the incumbent president's approval rating is the best way we can capture Harris' vulnerability to a lack of support from the large Arab-American community in Dearborn, Michigan over the US backing of Israel amid the war in Gaza. It also allows us to capture the limited upside to her candidacy from the CHIPS Act and Inflation Reduction Act, which have led to outsize investments in semiconductors, electric vehicles, and clean energy in Michigan. Despite these two signature pieces of fiscal legislation, approval of Biden and Harris still seems permanently damaged by the high inflation of 2021 and 2022.

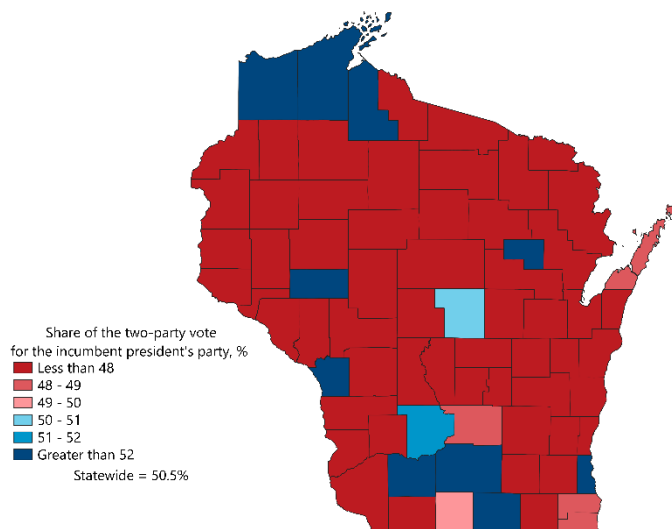
Limited downside for Harris in Wisconsin

Of the three northern swing states, Harris is projected to do best in Wisconsin, capturing 50.5% of the major-party vote (**Chart 10**). She does so by running large margins of victory in the state's two largest population centers of Milwaukee and Madison. Meanwhile, Trump doesn't garner enough votes, on net, in

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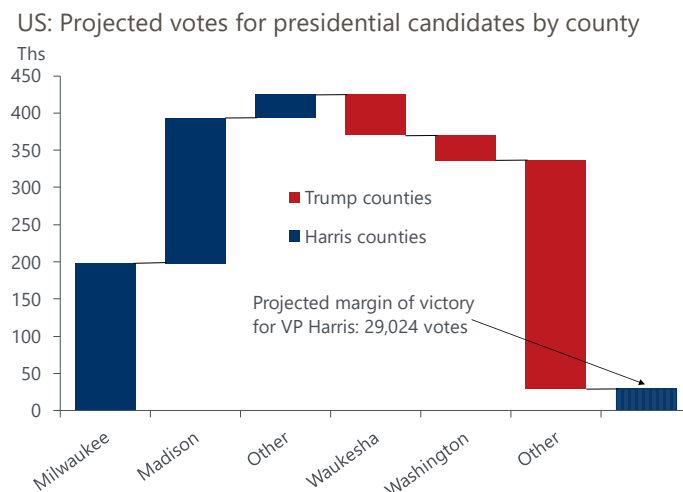
the WOW counties of Waukesha, Ozaukee, and Washington that reside in the Milwaukee metropolitan area, nor in other populous counties that include the cities of Green Bay, Racine, and Wausau (**Chart 11**).

Chart 10: How Wisconsin counties are projected to vote in 2024



Source: Oxford Economics

Chart 11: Harris runs up the score in Milwaukee and Madison



Source: Oxford Economics/Haver Analytics

Harris' projected victory in Wisconsin would be about 9,000 votes more than Biden's 2020 margin. Her weaknesses in the Pennsylvania and Michigan county model are not as much of an issue in the Badger State. Statewide, the unemployment rate is only forecasted to rise by 16bps in the two quarters ahead of the 2024 election, compared to 31bps in Pennsylvania. Moreover, we find that the incumbent president's approval does the best job in predicting county election outcomes in Wisconsin only when the election occurs during or immediately after a downturn, which is not the case currently.

On Tuesday, news had just broken that Harris had selected Minnesota Governor Tim Walz as her running mate. The literature suggests a VP candidate confers a small home-state advantage to the presidential ticket, but the models are agnostic as to whether Walz's high net-approval rating in Minnesota will spill over into neighboring Wisconsin.

What to expect in the next model update

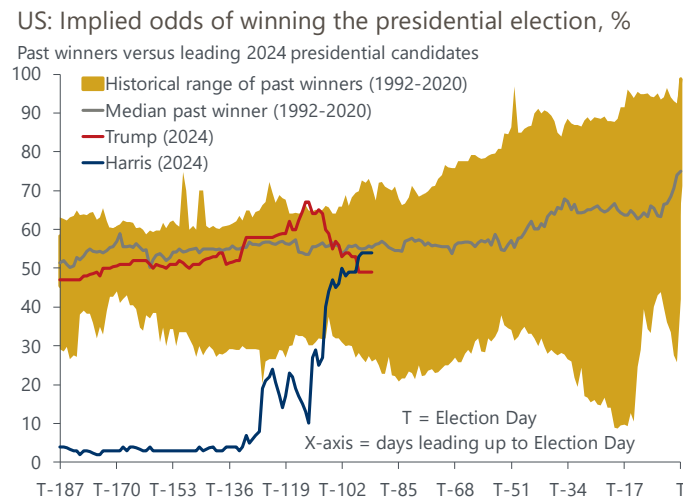
We will update our county election models for Pennsylvania, Michigan, and Wisconsin in October to incorporate the September vintage of our US cities and regions economic forecasts. The county election

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projections in this research briefing reflect the June vintage of our US subnational forecasts. Therefore, any changes to our baseline forecasts for unemployment and real disposable income at the national and state levels between the June and September vintages will impact the next update of our county election projections. As [Appendix 3](#) discusses, our county election forecasts for Pennsylvania and Michigan reflect recent voter registration activity. Any relative increase in registrations in blue or red counties during the next few months may have a noticeable impact on our updated projections this fall.

Political betting markets slightly favor Harris to win the election ([Chart 12](#)). Nevertheless, Harris' and Trump's odds of winning are still below the median probability of winning for past presidential election winners at comparable points in all election cycles since 1992. Therefore, this is still a wide-open race, likely with more twists and turns to come that will have an impact on voters' decisions in these and other battleground states.

Chart 12: Presidential race remains in flux



Source: Oxford Economics/Bloomberg/Iowa Electronic Markets/PredictIt

Appendix 1: How northern swing state counties will vote in November

Table 1: How Pennsylvania counties are projected to vote in 2024

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Adams	18,254	37,567	-19,313	16,803	37,696	-20,892
Allegheny	430,759	282,913	+147,846	388,523	245,222	+143,301
Armstrong	8,457	27,489	-19,032	6,869	24,890	-18,021
Beaver	38,122	54,759	-16,637	32,595	49,910	-17,314
Bedford	4,367	23,025	-18,658	3,538	20,583	-17,045
Berks	93,116	109,926	-16,810	80,976	105,559	-24,583
Blair	17,636	45,306	-27,670	14,469	40,255	-25,786
Bradford	8,046	21,600	-13,554	6,637	19,656	-13,020
Bucks	204,712	187,367	+17,345	184,088	174,265	+9,823

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County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Butler	37,508	74,359	-36,851	34,021	72,148	-38,127
Cambria	21,730	48,085	-26,355	18,296	45,382	-27,087
Cameron	634	1,771	-1,137	481	1,504	-1,022
Carbon	11,212	21,984	-10,772	9,508	20,519	-11,010
Centre	40,055	36,372	+3,683	33,751	33,039	+712
Chester	182,372	128,565	+53,807	165,813	126,577	+39,236
Clarion	4,678	14,578	-9,900	4,043	13,760	-9,717
Clearfield	9,673	29,203	-19,530	7,783	27,092	-19,309
Clinton	5,502	11,902	-6,400	4,608	10,813	-6,205
Columbia	10,532	20,098	-9,566	8,613	18,077	-9,465
Crawford	12,924	28,561	-15,637	10,821	26,058	-15,237
Cumberland	62,245	77,212	-14,967	57,371	74,809	-17,438
Dauphin	78,983	66,408	+12,575	73,176	62,700	+10,476
Delaware	206,709	118,639	+88,070	190,373	106,127	+84,246
Elk	4,522	12,140	-7,618	3,639	11,190	-7,551
Erie	68,286	66,869	+1,417	57,066	60,056	-2,990
Fayette	20,469	41,251	-20,782	15,722	34,332	-18,611
Forest	728	1,882	-1,154	597	1,825	-1,228
Franklin	22,422	57,245	-34,823	20,439	56,195	-35,756
Fulton	1,085	6,824	-5,739	829	6,098	-5,269
Greene	4,911	12,579	-7,668	3,875	11,440	-7,565
Huntingdon	5,445	17,061	-11,616	4,231	14,723	-10,492
Indiana	12,634	28,089	-15,455	10,347	25,414	-15,066
Jefferson	4,529	17,964	-13,435	3,462	15,709	-12,247
Juniata	2,253	9,649	-7,396	1,883	9,027	-7,144
Lackawanna	61,991	52,334	+9,657	54,677	46,738	+7,938
Lancaster	115,847	160,209	-44,362	102,827	154,907	-52,080

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County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Lawrence	15,978	29,597	-13,619	14,040	27,366	-13,326
Lebanon	23,932	46,731	-22,799	22,056	45,912	-23,855
Lehigh	98,498	84,418	+14,080	88,756	80,313	+8,444
Luzerne	64,873	86,929	-22,056	55,472	77,806	-22,334
Lycoming	16,971	41,462	-24,491	14,572	38,130	-23,558
McKean	5,098	14,083	-8,985	3,949	12,136	-8,187
Mercer	21,067	36,143	-15,076	17,881	33,046	-15,164
Mifflin	4,603	16,670	-12,067	3,729	14,698	-10,970
Monroe	44,060	38,726	+5,334	35,807	34,569	+1,237
Montgomery	319,511	185,460	+134,051	291,420	169,408	+122,013
Montour	3,771	5,844	-2,073	2,951	4,853	-1,902
Northampton	85,087	83,854	+1,233	74,559	77,958	-3,399
Northumberland	12,703	28,975	-16,272	10,824	26,506	-15,682
Perry	5,950	18,293	-12,343	5,275	17,703	-12,428
Philadelphia	604,175	132,870	+471,305	530,231	98,472	+431,759
Pike	13,052	19,241	-6,189	10,512	17,459	-6,946
Potter	1,726	7,239	-5,513	1,365	6,645	-5,279
Schuylkill	20,727	48,871	-28,144	18,272	46,301	-28,029
Snyder	4,910	13,983	-9,073	4,160	13,174	-9,015
Somerset	8,654	31,466	-22,812	7,201	29,081	-21,881
Sullivan	921	2,619	-1,698	802	2,551	-1,749
Susquehanna	6,236	15,207	-8,971	5,358	14,264	-8,906
Tioga	4,955	15,742	-10,787	4,031	14,056	-10,025
Union	7,475	12,356	-4,881	6,664	12,187	-5,523
Venango	7,585	18,569	-10,984	6,308	16,517	-10,209
Warren	6,066	14,237	-8,171	4,698	12,249	-7,550
Washington	45,088	72,080	-26,992	36,784	63,096	-26,312

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County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Wayne	9,191	18,637	-9,446	8,108	18,329	-10,221
Westmoreland	72,192	130,299	-58,107	63,216	120,727	-57,511
Wyoming	4,704	9,936	-5,232	3,925	9,155	-5,230
York	88,114	146,733	-58,619	78,072	146,276	-68,204
Total	3,461,221	3,379,055	+82,166	3,063,746	3,105,235	-41,489

Source: Oxford Economics

Table 2: How Michigan counties are projected to vote in 2024

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Alcona	2,142	4,848	-2,706	2,381	6,439	-4,059
Alger	2,053	3,014	-961	2,332	3,781	-1,449
Allegan	24,449	41,392	-16,943	28,116	52,102	-23,986
Alpena	6,000	10,686	-4,686	6,786	13,176	-6,390
Antrim	5,960	9,748	-3,788	6,846	12,430	-5,584
Arenac	2,774	5,928	-3,154	3,168	7,306	-4,138
Baraga	1,478	2,512	-1,034	1,748	3,319	-1,571
Barry	11,797	23,471	-11,674	13,239	29,789	-16,550
Bay	26,151	33,125	-6,974	29,493	39,918	-10,426
Benzie	5,480	6,601	-1,121	5,983	8,229	-2,246
Berrien	37,438	43,519	-6,081	43,231	55,478	-12,247
Branch	6,159	14,064	-7,905	7,212	18,456	-11,244
Calhoun	28,877	36,221	-7,344	33,795	45,988	-12,193
Cass	9,130	16,699	-7,569	10,668	22,393	-11,726
Charlevoix	6,939	9,841	-2,902	7,514	12,014	-4,500
Cheboygan	5,437	10,186	-4,749	5,894	12,847	-6,953
Chippewa	6,648	10,681	-4,033	7,967	13,811	-5,845
Clare	5,199	10,861	-5,662	6,126	13,970	-7,844

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County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Clinton	21,968	25,098	-3,130	24,635	30,399	-5,764
Crawford	2,672	5,087	-2,415	3,095	6,662	-3,567
Delta	7,606	13,207	-5,601	8,875	17,256	-8,381
Dickinson	4,744	9,617	-4,873	5,583	12,592	-7,009
Eaton	31,299	31,798	-499	35,359	37,955	-2,596
Emmet	9,662	12,135	-2,473	10,467	14,853	-4,386
Genesee	119,390	98,714	+20,676	137,895	120,778	+17,117
Gladwin	4,524	9,893	-5,369	5,116	12,380	-7,264
Gogebic	3,570	4,600	-1,030	4,175	6,172	-1,997
Grand Traverse	28,683	30,502	-1,819	31,471	37,010	-5,539
Gratiot	6,693	12,102	-5,409	7,582	15,226	-7,644
Hillsdale	5,883	17,037	-11,154	6,822	22,025	-15,204
Houghton	7,750	10,378	-2,628	8,725	12,938	-4,213
Huron	5,490	12,731	-7,241	5,922	15,642	-9,720
Ingham	94,212	47,639	+46,573	107,421	53,148	+54,273
Ionia	10,901	20,657	-9,756	12,582	25,529	-12,946
Iosco	5,373	9,759	-4,386	6,028	12,090	-6,062
Iron	2,493	4,216	-1,723	2,932	5,619	-2,687
Isabella	14,072	14,815	-743	16,684	19,381	-2,697
Jackson	31,995	47,372	-15,377	36,698	58,712	-22,014
Kalamazoo	83,686	56,823	+26,863	92,662	67,105	+25,557
Kalkaska	3,002	7,436	-4,434	3,496	9,893	-6,397
Kent	187,915	165,741	+22,174	212,626	203,417	+9,209
Keweenaw	672	862	-190	774	1,085	-311
Lake	2,288	3,946	-1,658	2,763	5,461	-2,698
Lapeer	16,367	35,482	-19,115	17,957	43,791	-25,835
Leelanau	8,795	7,916	+879	9,689	9,726	-37

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County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Lenawee	20,918	31,541	-10,623	23,647	39,316	-15,669
Livingston	48,220	76,982	-28,762	53,640	92,344	-38,705
Luce	842	2,109	-1,267	956	2,693	-1,737
Mackinac	2,632	4,304	-1,672	3,012	5,476	-2,463
Macomb	223,952	263,863	-39,911	262,002	315,968	-53,966
Manistee	6,107	8,321	-2,214	6,864	10,591	-3,728
Marquette	20,465	16,286	+4,179	23,064	20,071	+2,993
Mason	6,802	10,207	-3,405	7,682	12,913	-5,230
Mecosta	7,375	13,267	-5,892	8,335	16,968	-8,633
Menominee	4,316	8,117	-3,801	5,138	10,991	-5,854
Midland	20,493	27,675	-7,182	22,997	33,634	-10,638
Missaukee	1,967	6,648	-4,681	2,201	8,494	-6,293
Monroe	32,975	52,710	-19,735	39,327	66,370	-27,043
Montcalm	9,703	21,815	-12,112	10,970	28,802	-17,832
Montmorency	1,628	4,171	-2,543	1,872	5,513	-3,641
Muskegon	45,643	45,133	+510	52,324	57,146	-4,822
Newaygo	7,873	18,857	-10,984	8,982	24,555	-15,572
Oakland	434,148	325,971	+108,177	498,178	374,470	+123,708
Oceana	4,944	8,892	-3,948	5,676	11,548	-5,871
Ogemaw	3,475	8,253	-4,778	3,932	10,579	-6,647
Ontonagon	1,391	2,358	-967	1,479	3,063	-1,584
Osceola	3,214	8,928	-5,714	3,720	11,567	-7,847
Oscoda	1,342	3,466	-2,124	1,522	4,466	-2,944
Otsego	4,743	9,779	-5,036	5,410	12,679	-7,270
Ottawa	64,705	100,913	-36,208	73,705	126,248	-52,544
Presque Isle	2,911	5,342	-2,431	3,197	6,867	-3,671
Roscommon	5,166	9,670	-4,504	6,118	12,326	-6,207

Predicting how swing voters in battleground counties will vote

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Saginaw	51,088	50,785	+303	59,430	61,169	-1,739
Sanilac	5,966	16,194	-10,228	6,597	19,841	-13,245
Schoolcraft	1,589	3,090	-1,501	1,839	4,065	-2,225
Shiawassee	15,347	23,149	-7,802	17,362	28,526	-11,164
St. Clair	31,363	59,185	-27,822	35,557	74,058	-38,501
St. Joseph	9,262	18,127	-8,865	10,941	24,155	-13,214
Tuscola	8,712	20,297	-11,585	9,587	24,539	-14,952
Van Buren	16,803	21,591	-4,788	19,409	27,779	-8,371
Washtenaw	157,136	56,241	+100,895	184,160	63,998	+120,162
Wayne	597,170	264,553	+332,617	713,908	300,286	+413,622
Wexford	5,838	12,102	-6,264	6,589	15,543	-8,954
Total	2,804,040	2,649,852	+154,188	3,241,859	3,217,912	+23,948

Source: Oxford Economics

Table 3: How Wisconsin counties are projected to vote in 2024

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Adams	4,329	7,362	-3,033	3,983	7,095	-3,112
Ashland	4,801	3,841	+960	4,517	3,870	+647
Barron	9,194	15,803	-6,609	8,273	15,406	-7,133
Bayfield	6,147	4,617	+1,530	6,009	4,901	+1,107
Brown	65,511	75,871	-10,360	65,058	75,854	-10,797
Buffalo	2,860	4,834	-1,974	2,516	4,759	-2,243
Burnett	3,569	6,462	-2,893	3,425	6,478	-3,053
Calumet	12,116	18,156	-6,040	12,103	18,486	-6,384
Chippewa	13,983	21,317	-7,334	13,219	20,900	-7,681
Clark	4,524	10,002	-5,478	4,036	9,882	-5,846
Columbia	16,410	16,927	-517	15,416	16,398	-981

Predicting how swing voters in battleground counties will vote

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Crawford	3,953	4,620	-667	3,546	4,391	-845
Dane	260,121	78,794	+181,327	269,485	73,764	+195,722
Dodge	16,356	31,355	-14,999	15,269	29,278	-14,009
Door	10,044	9,752	+292	9,701	10,243	-542
Douglas	13,218	10,923	+2,295	12,629	10,686	+1,943
Dunn	9,897	13,173	-3,276	9,457	13,741	-4,285
Eau Claire	31,620	25,341	+6,279	31,453	26,123	+5,329
Florence	781	2,133	-1,352	708	2,123	-1,414
Fond du Lac	20,588	35,754	-15,166	19,971	34,844	-14,873
Forest	1,721	3,285	-1,564	1,492	3,107	-1,615
Grant	10,998	14,142	-3,144	9,839	14,039	-4,200
Green	10,851	10,169	+682	10,019	10,134	-115
Green Lake	3,344	7,168	-3,824	3,159	7,003	-3,844
Iowa	7,828	5,909	+1,919	7,562	5,562	+2,001
Iron	1,533	2,438	-905	1,351	2,351	-1,000
Jackson	4,256	5,791	-1,535	3,762	5,628	-1,866
Jefferson	19,904	27,208	-7,304	18,888	26,976	-8,088
Juneau	4,746	8,749	-4,003	4,194	8,128	-3,934
Kenosha	42,193	44,972	-2,779	39,670	41,819	-2,150
Kewaunee	3,976	7,927	-3,951	3,711	8,035	-4,323
La Crosse	37,846	28,684	+9,162	37,584	28,625	+8,959
Lafayette	3,647	4,821	-1,174	3,513	4,596	-1,082
Langlade	3,704	7,330	-3,626	3,388	7,019	-3,631
Lincoln	6,261	10,017	-3,756	5,704	9,906	-4,201
Manitowoc	16,818	27,218	-10,400	15,922	26,948	-11,026
Marathon	30,808	44,624	-13,816	29,073	44,429	-15,356
Marinette	7,366	15,304	-7,938	6,734	14,985	-8,251

Predicting how swing voters in battleground counties will vote

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Marquette	3,239	5,719	-2,480	2,956	5,561	-2,606
Menominee	1,303	278	+1,025	1,205	220	+985
Milwaukee	317,527	134,482	+183,045	316,284	118,463	+197,821
Monroe	8,433	13,775	-5,342	7,645	13,117	-5,472
Oconto	6,715	16,226	-9,511	6,048	16,226	-10,178
Oneida	10,105	13,671	-3,566	9,412	13,677	-4,265
Outagamie	47,667	58,385	-10,718	45,418	58,653	-13,235
Ozaukee	26,517	33,912	-7,395	26,602	34,186	-7,584
Pepin	1,489	2,584	-1,095	1,399	2,515	-1,117
Pierce	9,796	12,815	-3,019	9,258	13,334	-4,076
Polk	9,370	16,611	-7,241	8,795	16,270	-7,475
Portage	20,428	19,299	+1,129	19,956	19,320	+636
Price	3,032	5,394	-2,362	2,715	5,354	-2,639
Racine	50,159	54,479	-4,320	49,298	53,232	-3,934
Richland	3,995	4,871	-876	3,660	4,788	-1,128
Rock	46,658	37,138	9,520	45,281	36,495	8,786
Rusk	2,517	5,257	-2,740	2,176	5,083	-2,907
Sauk	18,108	17,493	+615	19,709	18,586	+1,123
Sawyer	4,498	5,909	-1,411	5,504	7,839	-2,336
Shawano	7,131	15,173	-8,042	4,458	10,072	-5,614
Sheboygan	27,101	37,609	-10,508	16,867	23,659	-6,792
St. Croix	23,190	32,199	-9,009	25,007	37,476	-12,468
Taylor	2,693	7,657	-4,964	2,314	7,388	-5,075
Trempealeau	6,285	8,833	-2,548	5,546	8,721	-3,175
Vernon	7,457	8,218	-761	6,885	8,200	-1,315
Vilas	5,903	9,261	-3,358	5,728	9,528	-3,801
Walworth	22,789	33,851	-11,062	21,507	33,415	-11,908

Predicting how swing voters in battleground counties will vote

County	Actual 2020 results			Projected 2024 results		
	Joe Biden	Donald Trump	Margin of victory/defeat for Biden (+/-)	Kamala Harris	Donald Trump	Margin of victory/defeat for Harris (+/-)
Washburn	3,867	6,334	-2,467	3,605	6,430	-2,825
Washington	26,650	60,237	-33,587	26,136	60,373	-34,236
Waukesha	103,906	159,649	-55,743	104,344	159,024	-54,680
Waupaca	9,703	18,952	-9,249	8,757	17,977	-9,220
Waushara	4,388	9,016	-4,628	3,966	8,714	-4,748
Winnebago	44,060	47,796	-3,736	42,818	47,674	-4,856
Wood	16,365	24,308	-7,943	15,006	23,501	-8,495
Total	1,630,866	1,610,184	+20,682	1,592,607	1,563,583	+29,024

Source: Oxford Economics

Appendix 2: Backtesting our county models

Our county models accurately predict the winner of Pennsylvania and Michigan in all presidential elections going back to the 1992 Clinton-HW Bush contest (**Tables 4-5**).

Table 4: Track record of the Pennsylvania county model

Election year	Actual			Predicted		
	Incumbent party	Nonincumbent party	Winner	Incumbent party	Nonincumbent party	Winner
1992	1,791,841	2,239,164	Democrat	1,904,644	2,126,361	Democrat
1996	2,215,819	1,801,169	Democrat	2,419,575	1,903,728	Democrat
2000	2,485,967	2,281,127	Democrat	2,465,568	2,461,948	Democrat
2004	2,793,847	2,938,095	Democrat	2,533,525	2,766,082	Democrat
2008	2,655,885	3,276,363	Democrat	2,541,910	3,218,014	Democrat
2012	2,990,274	2,682,855	Democrat	3,014,358	2,530,679	Democrat
2016	2,926,458	2,970,742	Republican	2,778,460	3,159,175	Republican
2020	3,379,055	3,461,221	Democrat	3,421,097	3,580,380	Democrat

Source: Oxford Economics

Predicting how swing voters in battleground counties will vote

Table 5: Track record of the Michigan county model

Election year	Actual			Predicted		
	Incumbent party	Nonincumbent party	Winner	Incumbent party	Nonincumbent party	Winner
1992	1,554,940	1,871,182	Democrat	1,576,269	1,849,853	Democrat
1996	1,989,653	1,481,212	Democrat	2,156,991	1,772,168	Democrat
2000	2,170,418	1,953,139	Democrat	1,986,126	1,969,083	Democrat
2004	2,313,746	2,479,183	Democrat	2,144,083	2,255,840	Democrat
2008	2,048,639	2,872,579	Democrat	2,140,168	2,807,381	Democrat
2012	2,564,569	2,115,256	Democrat	2,801,706	2,029,288	Democrat
2016	2,268,839	2,279,543	Republican	2,268,427	2,469,756	Republican
2020	2,649,852	2,804,040	Democrat	2,561,801	2,628,005	Democrat

Source: Oxford Economics

The Wisconsin model is the least accurate, though (**Table 6**). In 2000, Vice President Al Gore carried Wisconsin by less than 6,000 votes, whereas the model predicts a winning margin of less than 8,000 for then Governor George W. Bush. Four years later, Senator John Kerry won Wisconsin by less than 12,000 votes. The model misses this narrow victory for the Democratic challenger in 2004. Wisconsin has a penchant for being a nailbiter. In four out of the past six presidential elections, the winning margin has been around 20,000 votes or less, making the Badger State a tough one to get right.

Table 6: Track record of the Wisconsin county model

Election year	Actual			Predicted		
	Incumbent party	Nonincumbent party	Winner	Incumbent party	Nonincumbent party	Winner
1992	930,855	1,041,066	Democrat	888,779	1,083,142	Democrat
1996	1,071,971	845,029	Democrat	1,232,625	1,067,060	Democrat
2000	1,242,987	1,237,279	Democrat	1,178,127	1,185,979	Republican
2004	1,478,120	1,489,504	Democrat	1,343,449	1,282,313	Republican
2008	1,262,393	1,677,211	Democrat	1,333,949	1,609,526	Democrat
2012	1,620,985	1,407,966	Democrat	1,692,459	1,233,674	Democrat
2016	1,382,536	1,405,284	Republican	1,378,100	1,615,351	Republican
2020	1,610,184	1,630,866	Democrat	1,614,470	1,624,599	Democrat

Source: Oxford Economics

Appendix 3: How we forecast presidential elections at the county level

There are two sets of county models that are used to predict the margin of victory or defeat for the incumbent party candidate. All are estimated as pooled regressions. The first set of regressions predict the share of the two-party vote that goes to the incumbent president's party (**Tables 7-9**). The historical sample of this first group of models include all presidential elections from the Clinton-HW Bush contest onward; limitations to county economic data prevented us from having a longer sample.

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Table 7: Forecasting the incumbent party candidate's share of the major-party vote in Pennsylvania

	Coefficient	Standard error	T-statistic
Constant	-0.0150	0.0073	-2.0534
Real disposable income per household, 1-year % change	0.0014	0.0004	3.4680
Unemployment rate when election occurs during or immediately after a recession, 2-quarter ppt change	-0.0037	0.0014	-2.5231
Unemployment rate when election does not occur with a recessionary backdrop, 2-quarter ppt change	-0.0193	0.0038	-5.0435
Fatigue dummy when incumbent is a Democrat	-0.0493	0.0042	-11.8540
Fatigue dummy when incumbent is a Republican	-0.0376	0.0038	-9.8487
Incumbent party's two-party vote share in prior election	*County fixed effects		
<i>*Independent coefficient for each state; all near 1 and highly significant</i>			
Adjusted R-squared	0.9436		

Source: Oxford Economics

Table 8: Forecasting the incumbent party candidate's share of the major-party vote in Michigan

	Coefficient	Standard error	T-statistic
Constant	0.0208	0.0095	2.1889
Real disposable income per household, 2-year % change	0.0013	0.0004	3.6317
Unemployment rate, 2-quarter ppt change	-0.0080	0.0012	-6.8163
Presidential approval rating, 1-year ppt change	0.0013	0.0002	8.3589
Democratic-incumbent dummy	-0.0396	0.0040	-10.0023
Fatigue dummy	-0.0640	0.0029	-22.0964
Incumbent party's two-party vote share in prior election	*County fixed effects		
<i>*Independent coefficient for each state; all near 1 and highly significant</i>			
R-squared	0.8985		

Source: Oxford Economics

Table 9: Forecasting the incumbent party candidate's share of the major-party vote in Wisconsin

	Coefficient	Standard error	T-statistic
Constant	0.0021	0.0111	0.1936
Real disposable income per household, 2-year % change	0.0022	0.0006	3.8669
Unemployment rate when election does not occur with a recessionary backdrop, 2-quarter ppt change	-0.0053	0.0060	-0.8798
Net presidential approval rating when election occurs during or immediately after a recession	0.0025	0.0007	3.6794
Fatigue dummy when incumbent is a Democrat	-0.0719	0.0061	-11.7748
Fatigue dummy when incumbent is a Republican	-0.0039	0.0112	-0.3464
Trump as challenger dummy	-0.0127	0.0076	-1.6786
Incumbent party's two-party vote share in prior election	*County fixed effects		
*Independent coefficient for each state; all near 1 and highly significant			
R-squared	0.8639		

Source: Oxford Economics

We then forecast turnout as a share of total voter registrations, or the voting-age population in the case of Wisconsin (**Tables 10-12**). Because of limitations to county registration data, our sample for this second set of regressions only traces back to the 1996 Clinton-Dole contest. We use economic and political variables to predict turnout.

The faster the rise in unemployment, the higher the turnout – presumably a function of voter dissatisfaction with the incumbent party. Voter fatigue is associated with higher turnout. We control for county fixed

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effects in the form of turnout in the prior presidential election and include a Trump dummy variable, whose coefficient increases in magnitude the more a county skews Republican. This dummy variable is meant to capture Trump's outperformance vis-à-vis past Republican candidates in many red counties within these traditional factory belt states.

Table 10: Forecasting turnout as a share of total registrations in Pennsylvania

	Coefficient	Standard error	T-statistic
Constant	0.5191	0.0312	16.6106
Unemployment rate, 2-quarter ppt change	0.0183	0.0015	11.9455
Trump dummy*Republican share of total registered voters	0.0816	0.0096	8.4778
Turnout rate in prior election	*County fixed effects		
<i>*Independent coefficient for each state; all highly significant</i>			
R-squared	0.6324		

Source: Oxford Economics

Table 11: Forecasting turnout as a share of total registrations in Michigan

	Coefficient	Standard error	T-statistic
Constant	0.2703	0.0211	12.8145
Trump dummy*Republican share of major-party vote in prior election	0.1369	0.0078	17.4430
Fatigue dummy	0.0089	0.0039	2.2582
Turnout rate in prior election	*County fixed effects		
<i>*Independent coefficient for each state; all highly significant</i>			
R-squared	0.6656		

Source: Oxford Economics

Table 12: Forecasting turnout as a share of the voting-age population in Wisconsin

	Coefficient	Standard error	T-statistic
Constant	0.3883	0.0357	10.8757
Unemployment rate, 2-quarter ppt change	0.0203	0.0044	4.6626
Turnout rate in prior election	*County fixed effects		
*Independent coefficient for each state; all highly significant			
R-squared	0.5264		

Source: Oxford Economics

Once we estimate turnout rates, we then apply them to the most recent voter registration numbers, which are available on the election websites of Pennsylvania and Michigan, and our voting-age population forecasts for 2024 in the case of Wisconsin. Therefore, we can estimate the total number of registered voters that show up at the ballot box in early November. With our estimates from the first set of regressions, we calculate how many ballots will be cast for the incumbent and nonincumbent party candidates.